



DFM / DFA Analysis

boss-plate

ALUMINIUM A356 (CAST) · LOW-PRESSURE DIE CASTING

GENERATED 2026-08-11 · 8/11 RULES EVALUATED

EXECUTIVE SUMMARY**Low-Pressure Die Casting · Aluminium A356 (cast)**

Judged against the Low-pressure die casting ruleset — 8 of 11 rules could be evaluated on this geometry.

58

MANUFACTURABILITY

3

FINDINGS

2

HIGH SEVERITY

none priced

PRICED / YEAR

THE PART**No 3D view was captured**

The report was generated without an open 3D view. The analysis below is unaffected — it is measured from the solid, not from the picture.

ENVELOPE

40 x 40 x 20 mm

VOLUME

14.647 cm3

SURFACE

54.036 cm2

FACES

9

VIEW

—

WHAT IS WRONG, WORST FIRST

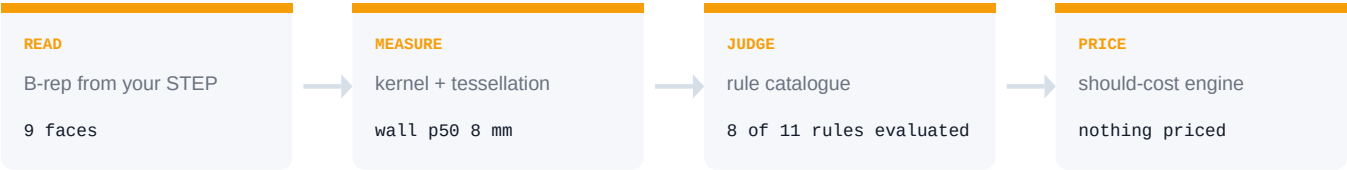
	FINDING	MEASURED	GUIDELINE	IMPACT / YR
HIGH	Wall area below the minimum permanent-mould draft	100 %	<= 5 %	not priced
HIGH	Tolerance tighter than low-pressure die casting holds	0.2 x	>= 1 x	not priced
MEDIUM	Cored holes below the core draft	0 deg	>= 2 deg	not priced

RULE COVERAGE 72.7% — 8 of 11 evaluated, 3 could not be measured on this geometry and are NOT passes.

ANALYSIS LIMITS

Drawing vs model: "Ø6.8 ±0.05" (6.8 mm) does not match the model's hole diameter of 6 mm — 0.8 mm apart, outside its own tolerance band. One of the two documents is wrong, and this tool cannot say which.

How this report was produced, and from what



MEASURED GEOMETRY	
BOUNDING BOX	VOLUME
40 x 40 x 20 mm	14.647 cm3
WALL p5 / p50 / p95	WALL UNIFORMITY
4.96 / 8 / 8 mm	moderate
DRAW DIRECTION	UNDERCUT REGIONS
[0, 0, 1]	0
WALL AREA BELOW MIN DRAFT	UNCLASSIFIED AREA
100 %	0 %
HOLES & CUT-OUTS	INTERNAL CUT LENGTH
1 (1 round, 0 shaped)	18.9 mm

Cross-check: 2V/A gives 5.42 mm against the ray-cast median of 8 mm, from rays covering 69.3% of the surface. 2V/A, exact for a thin uniform shell and indicative otherwise — an independent check on the ray cast

WHAT NADCA #402 PROMISES FOR THIS PART

±0.275 mm	+0.14 mm	0.2 mm	0.2×
LINEAR AT 40 MM · STANDARD	PARTING-LINE ADDER (16 CM²)	FLATNESS AT 57 MM DIAG	WORST PMI BAND (6 MM)

From NADCA Product Specification Standards for Die Castings (2021, 11th ed.), read first-hand: linear capability from Tables S/P-4A-1 at this part's largest dimension, the PLUS-ONLY parting-line adder from S/P-4A-2 at the bounding footprint perpendicular to the draw (an upper bound on true projected area, so the adder is an upper bound too), flatness from S/P-4A-8 at the bounding diagonal. A drawing tighter than these numbers is not impossible — it is a conversation with the die caster, and a cost.

WHAT ISO 8062-4 PROMISES FOR THIS CASTING

0.5 mm	0.2×	2.8°
S8 TOTAL BAND · PERMANENT MOULD	WORST PMI BAND (6 MM)	DRAFT, TABLE 6 AT 20 MM DRAW

From ISO 8062-4:2017, read first-hand: dimensional capability from Table 2 at the grade Annex B.1 selects for this metal group by metallic permanent mould, judged at the loosest of the printed band; draft from Table 6 at the part's own draw extent, Grade A (fine) external — the least demanding printed column, so internal walls need more. Steel, nickel and cobalt print no permanent-mould column, and this panel abstains for them.

WHAT THE AI READ OFF THE 2D DRAWING

5	1	3 1x
DIMENSIONS (5 TOLERANCED)	GD&T FRAMES	CONFIRMED / CONFLICTS VS 3D

CONFLICT — "Ø6.8 ±0.05": the model's hole diameter measures 6 mm, 0.8 mm from the drawing. One of the two documents is wrong; this tool cannot say which.

Drawing DWG-BOSS-E2E rev A · units mm · readability good · general tolerance note "ISO 2768-mK". Extracted by AI vision and reviewed before judging; every extracted value was then judged by the same deterministic engines as a typed input, and each finding built on one is labelled drawing-read. The full row-by-row extraction with verbatim callouts is in the Excel export.

ACTIONS

3 decisions, worst first

FINDING, AND THE CHANGE THAT CLEARS IT		OWNER (ROLE) / DECISION NEEDED	
1	Wall area below the minimum permanent-mould draft Add at least 1° of draft to every drawn wall, 2-3° on deep or cosmetic faces. 100 % of wall under 2.8° -> <= 5 % of wall under 2.8°	Design + Tooling Decide without a price — No modelled cost driver connects this finding to the piece-price engines	
2	Tolerance tighter than low-pressure die casting holds Open the tolerance, or add machining stock on that feature and quote the operation. 0.2 x ISO 8062-4 capability -> >= 1 x ISO 8062-4 capability	Cost engineering Decide without a price — No modelled cost driver connects this finding to the piece-price engines	
3	Cored holes below the core draft Take cored holes to 2 degrees per side. 0 deg per side -> >= 2 deg per side	Design engineering Decide without a price — No modelled cost driver connects this finding to the piece-price engines	
DUE			
BY OWNER			
Design + Tooling		1 action, 1 high severity	not priced
Cost engineering		1 action, 1 high severity	not priced
Design engineering		1 action	not priced

TOLERANCES

Read from the model

5 dimensions, 1 geometric tolerance and 0 datums were read from the model's semantic PMI. The tightest total band called out is 0.1 mm, and that is what the process-capability rules below are measured against.

tolerance – mm

3 findings · 8/11 rules evaluated

MANUFACTURABILITY SCORE

58 / 100

RULE COVERAGE

72.7%

PRICED IMPACT / YEAR

none priced

Wall area below the minimum permanent-mould draft

HIGH

MEASURED 100 % of wall under 2.8° · GUIDELINE <= 5 % of wall under 2.8°

The part shrinks onto a steel mould as it solidifies under pressure. Without taper it grips, and the ejectors either mark the face or tear it.

WHAT TO DO

Add at least 1° of draft to every drawn wall, 2-3° on deep or cosmetic faces.

COST IMPACT

Not priced. No modelled cost driver connects this finding to the piece-price engines.

SOURCE [INDUSTRY CONSENSUS, no primary source audited]: Permanent-mould design guidance: minimum draft about 1°, with 2-3° general...

THRESHOLD: process-generic — Aluminium A356 (cast) was applied, but this rule carries no alloy-specific band, so the process-wide...

Tolerance tighter than low-pressure die casting holds

HIGH

MEASURED 0.2 x ISO 8062-4 capability · GUIDELINE >= 1 x ISO 8062-4 capability

Low-pressure die casting is the most accurate of the permanent-mould routes — ISO 8062 CT6-CT7 against gravity's CT7-CT9 — but it is still a casting. A band tighter than about 0.6 mm total is a machining requirement, and the stock for it has to be on the part.

WHAT TO DO

Open the tolerance, or add machining stock on that feature and quote the operation.

COST IMPACT

Not priced. No modelled cost driver connects this finding to the piece-price engines.

SOURCE [NAMED STANDARD, not read first-hand]: ISO 8062-4:2017, READ FIRST-HAND (Tables 2 + B.1): general dimensional tolerance cap...

THRESHOLD: process-generic — Aluminium A356 (cast) was applied, but this rule carries no alloy-specific band, so the process-wide...

Cored holes below the core draft

MEDIUM

MEASURED 0 deg per side · GUIDELINE >= 2 deg per side

Low-pressure filling is gentler than HPDC but the casting still shrinks onto a steel core, and the pin sees the same binding risk over a longer cycle.

WHAT TO DO

Take cored holes to 2 degrees per side.

COST IMPACT

Not priced. No modelled cost driver connects this finding to the piece-price engines.

SOURCE [INDUSTRY CONSENSUS, no primary source audited]: Permanent-mould and low-pressure die design guidance: cored holes drafted...

THRESHOLD: process-generic — Aluminium A356 (cast) was applied, but this rule carries no alloy-specific band, so the process-wide...

PASSED (5)

- Local section below what a low-pressure fill reaches — measured 4.96 mm against >= 2 mm
- Non-uniform wall thickness — measured 0.38 (p95-p5)/p50 against <= 0.7 (p95-p5)/p50
- Undercuts need a slide or a sand core — measured 0 regions against <= 0 regions
- Hole too small to be cast — it will be drilled — measured 6 mm dia against >= 6 mm dia
- Cored hole beyond the core slenderness limit — measured 3.33 core L/D against <= 6 core L/D

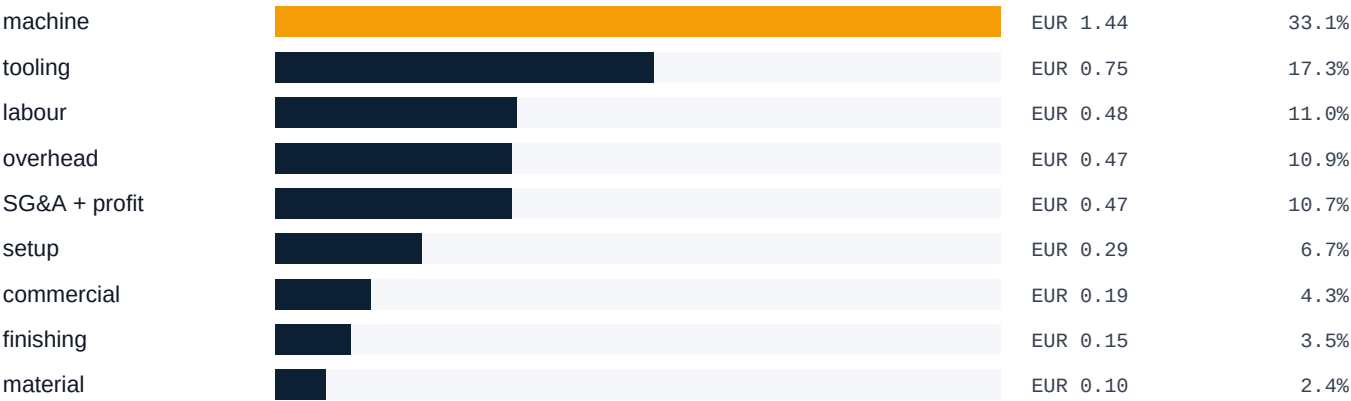
NOT EVALUATED (3) — these were NOT checked and are NOT passes

- Rib heavier than the wall it stands on — no measurement available for "maxRibThicknessToWall" on this geometry
- Rib too thin to fill — no measurement available for "minRibThicknessToWall" on this geometry
- Rib taller than it can be filled and drawn — no measurement available for "maxRibHeightToWall" on this geometry

COST DRIVERS

Where the money goes at Low-Pressure Die Casting

The largest single driver is machine time at EUR 1.44 per part — 33% of the EUR 4.35 piece price. That is the lever the findings below should be read against.



THE LEVERS BEHIND THE ROWS: buy-to-fly 0.05 kg in · 55 s cycle · 4% scrap · EUR 86,963 tooling investment
computeShouldCost on the chosen route — the same engine that prices the alternatives table, so the rows reconcile.

Your route is Low-Pressure Die Casting, against the other 27

The findings above are YOUR route and nobody else's. This page exists only to answer the next question — would another process make the same geometry for less. Each row is the same measured geometry run through THAT process's own rule family, priced by computeShouldCost and carbon-scored by computeCarbon on the cost engine's own input mass. Nothing is blended into a single ranking.

Process	Score	Rules	EUR/part	Tooling	kg CO2e
Cold Heading / Upsetting	80	5/5	EUR 0.29	EUR 12,196	0.21
-EUR 4.06/part vs your route, -EUR 74,766 tooling · 1 high-severity					
Forging (Cold)	67	5/5	EUR 0.32	EUR 19,570	0.23
-EUR 4.03/part vs your route, -EUR 67,393 tooling · 1 high-severity					
Extrusion	60	5/6	EUR 0.39	EUR 25,000	0.24
-EUR 3.96/part vs your route, -EUR 61,963 tooling · 2 high-severity					
Roll Forming	0	2/6	EUR 0.92	EUR 121,570	0.23
-EUR 3.43/part vs your route, +EUR 34,607 tooling · 2 high-severity · Only 2 of 6 rules could be evaluated, so this score...					
Broaching	SECONDA...	4/4	EUR 0.96	EUR 18,118	0.25
NOT A ROUTE: Broaching finishes a hole or a form in an existing part; it does not produce the part.					
Stamping / Deep Drawing	0	1/12	EUR 1.31	EUR 183,533	0.32
-EUR 3.04/part vs your route, +EUR 96,570 tooling · 1 high-severity · Only 1 of 12 rules could be evaluated, so this scor...					
Fine Blanking	100	1/6	EUR 1.52	EUR 61,178	0.28
-EUR 2.83/part vs your route, -EUR 25,785 tooling · Only 1 of 6 rules could be evaluated, so this score rests on a partia...					
Laser Cutting + Bending	0	1/12	EUR 2.00	EUR 8,000	0.25
-EUR 2.35/part vs your route, -EUR 78,963 tooling · 1 high-severity · Only 1 of 12 rules could be evaluated, so this scor...					
Deep Drawing (Multi-stage)	71	4/5	EUR 2.17	EUR 96,766	0.28
-EUR 2.18/part vs your route, +EUR 9,804 tooling · 1 high-severity					
Forging (Hot)	55	5/6	EUR 2.75	EUR 71,963	0.27
-EUR 1.60/part vs your route, -EUR 15,000 tooling · 2 high-severity					
Hydroforming	50	4/5	EUR 3.00	EUR 222,748	0.25
-EUR 1.35/part vs your route, +EUR 135,785 tooling · 2 high-severity					
Die Casting (Aluminium)	57	9/17	EUR 3.24	EUR 92,355	0.34
-EUR 1.11/part vs your route, +EUR 5,393 tooling · 2 high-severity · Only 9 of 17 rules could be evaluated, so this score...					
Gravity Die Casting	66	10/13	EUR 3.33	EUR 71,766	0.31
-EUR 1.02/part vs your route, -EUR 15,196 tooling · 2 high-severity					
Shell Mould Casting	55	5/5	EUR 3.35	EUR 32,785	0.33
-EUR 1.00/part vs your route, -EUR 54,178 tooling · 2 high-severity					
Vacuum-Assisted Die Casting	57	9/17	EUR 3.96	EUR 112,748	0.34
-EUR 0.39/part vs your route, +EUR 25,785 tooling · 2 high-severity · Only 9 of 17 rules could be evaluated, so this scor...					
Sand Casting	61	7/15	EUR 4.00	EUR 18,471	0.37
-EUR 0.35/part vs your route, -EUR 68,492 tooling · 2 high-severity · Only 7 of 15 rules could be evaluated, so this scor...					
Turning (CNC)	SECONDA...	3/5	EUR 4.11	EUR 1,500	0.26
NOT A ROUTE: Turning is a route only for a part that starts as bar or a turned blank. On a non-axisymmetric part it is a...					
Machining (CNC)	100	5/7	EUR 4.21	EUR 4,000	0.46
-EUR 0.14/part vs your route, -EUR 82,963 tooling					
Low-Pressure Die Casting	58	8/11	EUR 4.35	EUR 86,963	0.25
YOUR ROUTE — the findings above are this one · 2 high-severity					
Centrifugal Casting	NOT VIABLE	3/3	EUR 4.54	EUR 20,353	0.26
CANNOT MAKE THIS PART: Part is not a body of revolution — Cast an axisymmetric blank and machine the non-axisymmetric fea...					
Semi-Solid Casting (Thixo/Rheo)	57	5/6	EUR 5.43	EUR 122,944	0.31
+EUR 1.08/part vs your route, +EUR 35,981 tooling · 1 high-severity					
Metal Spinning	NOT VIABLE	3/3	EUR 6.42	EUR 6,098	0.33
CANNOT MAKE THIS PART: Part is not a body of revolution — Spin an axisymmetric shell and add the non-axisymmetric feature...					
Squeeze Casting	55	6/6	EUR 6.45	EUR 112,748	0.26
+EUR 2.10/part vs your route, +EUR 25,785 tooling · 2 high-severity					
Deep-Hole / Gun Drilling	SECONDA...	4/4	EUR 9.62	EUR 2,500	0.29
NOT A ROUTE: Gun drilling makes a deep bore in an existing part; it does not produce the part.					

Investment Casting	62	9/11	EUR 15.41	EUR 41,178	0.42
+EUR 11.06/part vs your route, -EUR 45,785 tooling · 2 high-severity					
Wire EDM	SECONDA...	3/4	EUR 17.82	EUR 800	0.39
NOT A ROUTE: Wire EDM cuts a profile from a blank somebody else made, and cuts it slowly. It is a toolroom and low-volume...					
Open-Die Forging	33	5/5	EUR 26.70	EUR 6,079	0.44
+EUR 22.35/part vs your route, -EUR 80,884 tooling · 3 high-severity					
Laser Powder Bed Fusion (DMLS/SL...	43	4/4	EUR 89.98	EUR 0	0.73
+EUR 85.63/part vs your route, -EUR 86,963 tooling · 2 high-severity					

Cold Heading / Upsetting prices 4.06 EUR/part below Low-Pressure Die Casting on lower tooling (EUR 12,196 against EUR 86,963), and scores 80 over 5 of 5 rules with 1 high-severity finding. That is a quotation to ask for, not a decision this tool has taken for you.

Not applicable to Aluminium A356 (cast): Die Casting (Zinc), Hot Stamping (Press Hardening), Lamination Stamping (Electrical Steel), Injection Moulding, Rotational Moulding, Composite Layup (RTM), Rubber Moulding (Compression/Injection), Thermoforming, Metal Injection Moulding (MIM), Powder Metallurgy (Press & Sinter).

Where every number came from

GEOMETRY	Measured by an OpenCascade kernel. Draft angles, undercut classification and wall thickness on the tessellation — which works on freeform surfaces, where a plane-and-cylinder analysis measures almost nothing on a real cast or moulded part. Holes, apertures and features from the B-rep topology. The AI wrote none of the numbers in this report.
DRAW DIRECTION	Chosen by sweeping candidate axes and scoring undercut area, not assumed. Undercuts are separated from zero-draft drag faces: a zero-draft wall is fixable with a degree of taper, an undercut buys a slide or a lifter, and reporting them as one number would overstate the tooling problem.
COST	Re-run through the same deterministic engines used elsewhere in BrainSpark, once with the geometry as drawn and once with the rule satisfied. Findings the engines cannot price say so, with the reason.
THRESHOLDS	Every DIMENSION here was measured from your file and is reproducible. The GUIDELINES they are compared against are not of the same standing. 207 of 248 rest on industry consensus — widely published and mutually consistent, but not audited against a primary standard and not validated against measured scrap data. 38 name a published standard that has not been read first-hand; 3 come from this tool's own model. Every finding carries its grade. Treat a finding as a screening result that opens a conversation with your supplier — not as a specification.
THREE OUTCOMES, NOT TWO	Failed, passed, and NOT EVALUATED. A rule whose measurement this part does not provide is listed as unevaluated with the reason, never as a pass — so the coverage figure beside the score tells you how much of the catalogue actually ran.
DFA	Handling times follow the published Boothroyd-Dewhurst METHOD. Their tables are copyrighted and are not reproduced; the coefficients here are ours and should be calibrated against your own line data before a labour commitment. Part symmetry is MEASURED by rotating each solid and intersecting it with itself, not inferred from inertia. The three DFA questions concern function and intent, which a static solid cannot answer — until every part is answered, the theoretical minimum and the index are withheld.
WHAT THIS DOES NOT COVER	Threads are not recognised. GD&T is read only where the file carries semantic AP242 PMI. Ribs are found from opposed planar side faces on a planar base, so a rib with curved sides is not checked. Surface finish and material come from your input, not from the model. The tool-reach sweep does not model the holder, the spindle nose or the machine envelope, so a face it calls reachable may still not be.